Overview of Ecosystem Service: Recreation

Nature-based recreation is valued on a capital-account basis: realized visits to high-quality natural sites, priced at what each traveler spends to make them. The headline sums the ground travel of residents and of tourists to the sites, following the source's own combined total. The accommodation tourists stay in and the international air travel that brings them to the country are valued as separate additive components and published beside the total, exactly as the source reports them. The base year is 2019.

Quantity

Sites. Six SEALS7 land-cover class shares plus the accessible protected-area share combine into a 0-3 environment class; urban share and distance-to-road combine into a 1-5 accessibility class; the two cross into a 1-9 site rank, of which rank 9 is a high-quality site. Contiguous high-quality pixels dissolve into discrete sites by 8-connected labelling, so one large park is one destination rather than many independent site pixels.

Trips. Each population pixel carries an annual trip budget: population times a participation level, split over distance bands within a fixed 50 km cutoff by a closed-form exponential decay. Participation and decay rate come per country from the demand-parameter table (his documented calibration: one worldwide group). A band's fixed budget divides evenly across the sites its ring reaches; a band with no reachable site, and all demand beyond the cutoff, is tracked as unmet rather than discarded. Residents travel from their home pixels (WorldPop) with the full annual rate; tourists travel from their accommodation, each person-night carrying the rate's daily share (a/365), built by allocating each country's UNWTO overnight stays over its national hotel stock, with the nearest reported year standing in where 2019 is unreported and a country with no overnights data in any year kept missing rather than zero. The workbook's Units column reads Thousands, so the panel's numbers are read as thousands of person-nights, the same convention the arrivals sheet already gets.

Price

Ground travel: realized visits price the round trip to their band's midpoint distance at the origin country's fuel cost: the 2019 per-litre gasoline price times the IEA GFEI fuel economy of 7.1 litres per hundred kilometres (the staged table's own cost-per-km column applies that economy per kilometre, one hundred times too high, so the price column is read instead). A country without fuel-price data keeps its visits and gets a missing value, never free travel. **Accommodation**: overnight stays times the hotel price per person per night from the author's hybrid price surface; a price gap is missing, not free, and each country's price-coverage share is published beside the value. **Air travel**: arrivals by origin region times the staged region-to-region round-trip 2019 airfare matrix (the author's hub-pair scrape deflated by the jet-fuel producer price index; this module reads it as a staged input and makes no network request).

Valuation

$$\mathrm{GEP}_i = V_i^{\mathrm{daily}} + V_i^{\mathrm{tourist}}$$

the source's combined total, plain addition, so a country whose fuel-cost or overnights gap makes either ground channel unknown has a missing total rather than a partial one presented as complete. Accommodation and air travel are published beside the total as separate additive columns with their coverage shares, as the source's own country table reports them. One row per country on the r250 list.

A tourist person-night carries the daily share of the annual participation rate, a/365 or about 0.28 visits per night at zero distance. This is the author's correction of the delivered construction, which priced each night at the full annual rate of 102.5 visits. The demand parameters are his documented calibration, one worldwide group (a 102.50 annual visits per capita, decay 0.167 per km) from the England, Tel Aviv and Beijing surveys; country-specific groups are the outstanding task his methods page names.

Implementation notes

The flow engine windows by country where the source windows by parameter group; the maths is pixel-local within the 50 km cutoff and country ids partition each group's pixels, so the two give the same flows, pinned by a test that computes both formulations on one toy grid. The source also computes a distance-to-site transform its engine never reads; that step is not ported.

Inputs

All paths and parameters come from `es_parameters.csv` under the `recreation` service key: the six land-cover share rasters, the protected-area share, distance-to-road, the road-length reference grid, WorldPop, the fuel-cost table, the UNWTO all-data workbook, the hotel-points vector, the demand-parameter table, the hotel-price surface, the region crosswalk and the staged airfare matrix, each with a source row beside it.